

Data Exploration and Preprocessing Template

Phase: Data Collection and Preprocessing Phase

Project Title: Food Consumer Behaviour Analysis

Team ID: PNT2022TMID01234

Date: 15 October 2024

Maximum Marks: 10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains 10,000+ consumer records across 16 core attributes detailing consumer demographic profiles (Age, Gender, Household Income, Family Size, Occupation), dining channel preferences (Dine-in, Food Delivery App, Takeaway), average meal spend (\$), order frequency per week, preferred cuisines (Asian, Fast Food, Italian, Mexican, Healthy/Vegan), delivery transit duration, discount reliance, and satisfaction ratings (1 to 5 scale).
Data Cleaning	Missing Values: Imputed numerical attributes (Income, Spend) using group median by age/occupation; categorized missing categorical fields as 'Unspecified'. Duplicate Removal: Identified and dropped 182 duplicate transaction logs based on composite key (CustomerID, OrderTimestamp). Outlier Treatment: Applied Interquartile Range (IQR) filtering on meal spend and delivery duration; capped extreme values at the 99th percentile (\$180 meal spend, 90 mins delivery).
Data Transformation	Age Cohort Binning: Binned continuous age values into standard demographic cohorts: 'Gen Z (18-24)', 'Young Millennial (25-34)', 'Older Millennial (35-44)', 'Gen X (45-54)', and 'Boomers (55+)'. Income Tiering: Segmented income into 'Low Income (<\$30k)', 'Middle Income (\$30k-\$75k)', 'Upper Middle (\$75k-\$120k)', and 'High Income (\$120k+)'. Calculated Columns: Created Average Order Value (AOV = TotalSpend / TotalOrders), Spend Elasticity Index, and Satisfaction Quotient.
Data Type Conversion	- Order Date/Time converted from string object to native <code>datetime64[ns]</code> . - Spend, Income, Delivery Fee, and Discount Percentage converted to <code>float64</code> . - Order Frequency, Age, and Family Size converted to <code>int64</code> . - Ordering Channel, Cuisine, and Gender converted to categorical factors (<code>category</code>).
Column Splitting and Merging	Location Splitting: Split composite 'Customer_Location' strings into discrete 'City' and 'State' columns for regional Tableau mapping. Cuisine & Category Merging: Merged secondary cuisine tags into standardized primary cuisine families (e.g., 'Pizza', 'Pasta' -> 'Italian'). Name Parsing: Masked personally identifiable customer full names into anonymized Customer Hash IDs to ensure privacy compliance.
Data Modeling	Designed an analytical Star Schema in Tableau: Fact Table: <code>Fact_FoodOrders</code> (OrderID, CustomerID, ChannelID, CuisineID, SpendAmount, DiscountApplied, DeliveryDurationMinutes, Rating). Dimension Tables: <code>Dim_Customer</code> (Demographics, IncomeTier, AgeCohort), <code>Dim_Channel</code> (ChannelType, PlatformFee), <code>Dim_Cuisine</code> (CuisineCategory, HealthyFlag), and <code>Dim_Date</code> (Year, Quarter, Month, DayOfWeek, PeakHourFlag).
Save Processed Data	The cleaned, validated, and transformed dataset was exported as <code>food_consumer_behavior_cleaned.csv</code> and loaded as a Tableau Data Extract (.hyper) file for high-performance visual dashboard querying and public cloud publishing.